REPOSITORY 05

Artificial Intelligence in the Knowledge Economy

Ide & Talamàs (2025), *Journal of Political Economy*

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github.com/johnbarraza/ai-05-ide

1. The paper and the agent's problem

Question

How do AI capability and autonomy reorganize knowledge work, output, and wages?

One mechanism

Workers attempt problems; solvers handle exceptions. AI changes matching, span of control, and surplus shares.

Technology $n(z) = \dfrac{1}{h(1-z)}$

Source: Accepted manuscript, Sections 3.1–3.2, equations (1)–(4).

Competitive firm's choice

For human worker z, choose solver $s \geq z$:

$$\max_{s \geq z} \Pi_2^{nA}(s, z) = n(z)[s - w(z)] - w(s).$$

Autonomous AI adds the alternatives

$$\Pi_1^{A} = z_{AI} - r, \quad \Pi_2^{tA} = n(z)[z_{AI} - w(z)] - r,$$

$$\Pi_2^{bA} = n(z_{AI})[s - r] - w(s).$$

Free entry: the chosen structure earns zero profit.

2. Main result: capability and autonomy are distinct

Maintained environment: $z \in [0,1]$ observable; g continuous and strictly positive; $x \sim U[0,1]$; $h < h_0$; at most two layers; identical compute abundant relative to human time; $z_{AI} \in [0,1)$.

Proposition 5 · autonomous AI

Let $B = \{z \le z_{AI} : w^*(z) > w(z)\}$ and $T = \{z \ge z_{AI} : w^*(z) > w(z)\}$. Then

$$B \ne \varnothing \iff z_{AI} > \bar{z}_{AI}, \quad \bar{z}_{AI} \in \operatorname{int} W,$$

$$T \ne \varnothing \quad \forall z_{AI} \in [0,1).$$

The top claim can fail for $h \ge h_0$ and does not include $z_{AI} = 1$.

Proposition 6 · non-autonomous AI

Unique efficient equilibrium, $r^\star = 0$. If $z_{AI} \le w(0)$, AI is unused. If $z_{AI} > w(0)$, only the bottom uses it as solver. In either case:

1. $Y^* > Y^\star$;
2. some $z > 0$ has $w^\star(z) \le w(z)$, strictly if active;
3. near zero, $w^\star \ge \max\{w, w^*\}$, strictly if active;
4. near one, $w^\star \le w^*$, strictly for $z \ne 1$.

Intuition: autonomy changes feasible roles; **capability sets both the bottom-winner and adoption thresholds**.

3. What I did: a three-type threshold audit

Let $L < M < H$, $a = z_{AI}$, and

$$n_L = \frac{1}{h(1 - z_L)}.$$

Autonomous AI can produce alone, so $r^* = a$. Zero profit for AI solving for L gives

$$n_L(a - w_L^A) - a = 0 \quad \Rightarrow \quad \boxed{w_L^A = a[1 - h(1 - z_L)]}.$$

Hence

$$\boxed{L \text{ wins} \iff a > \bar{a}_L = \frac{w_L^0}{1 - h(1 - z_L)}}.$$

Non-autonomous AI leaves compute idle, so $r^\star = 0$. Conditional on adoption,

$$n_L(a - w_L^N) = 0 \Rightarrow w_L^N = a,$$

and Proposition 6 requires $a > w(0)$. Therefore

$$\boxed{w_L^N - w_L^A = ah(1 - z_L) > 0}.$$

What the exercise isolates

Capability turns adoption and bottom gains on; autonomy changes AI's outside option and who captures surplus.

Source: Discrete accounting extension; symbolic checks in discrete_model.py.

4. Where I did not believe the AI

Insert authentic evidence

Add your own photo as `hand/derivation.jpg`, then recompile. The short copy guide is in `.me/semana_5/`.

The tempting AI answer was:

> *"Distributional effects are driven by autonomy, not capability."*

My verdict: incomplete

Prop. 5 requires $z_{AI} > \bar{z}_{AI}$ for autonomous bottom winners. Prop. 6 requires $z_{AI} > w(0)$ before non-autonomous AI is even used.

Autonomy determines the role set; capability determines whether the relevant regime activates. The paper has two dimensions, not one.

Source: Accepted manuscript, Propositions 5–6, pp. 25–28.